Write code logic for the given Scenarios:

1. A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?

Steps:

1. Data Collecting - Collect the relevant data from the startup.

2. Data Preprocessing - Cleaning, handling missing data

3. Feature Selection - Remove highly correlated Features. Then use RFE or SelectKBest and find best Features needed.

4. Model Evaluation - Train different models using selected features.

5. Best Model Selection - Evaluate using Accuracy, ROC-AUC, and F1-Score and arrive at best model

---

2. You're creating a new project for managing student data. The project name is student_portal. What is the step-by-step command to start it?write logic for this django project

Django Command :- django-admin startproject student_portal

Steps:

1. Open command prompt or terminal

2. Go to the folder where you want the project

3. Run Django command specified above

4. This creates a folder named student_portal with required django files.

---

3. A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?

Steps:

1. Data Preprocessing - Cleaning, handling missing data and scale data.

2. Feature Reduction - Remove highly correlated Features. Then use RFE or SelectKBest and find best Features needed. Alternatively you can also apply PCA to reduce Dimensionality

3. Model Evaluation - Train different models using selected features.

4. Best Model Selection - Evaluate using Accuracy or RMSE based on classification or regression model and arrive at best model

---

4. A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?

Steps:

1. Data Collection - Collect data containing user ratings, user id and book id

2. User-Item matrix - make a User-Item matrix with ratings as values.

3. Similarity matrix - Make a User similarity matrix or Items similarity matrix using Cosine Similarity.

4. Unique books - Identify the books, based on similar users, unread by the target user and Recommend this.

---

5. A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?

Steps:

1. Data Collecting - Collect the relevant data from the bank.

2. Data Preprocessing - Cleaning, handling missing data and scale data.

3. Feature Selection - Remove highly correlated Features. Then use RFE or SelectKBest and find best Features needed. Alternatively you can also apply PCA to reduce Dimensionality

4. Model Evaluation - Train different models using selected features.

5. Best Model Selection - Evaluate using Accuracy, ROC-AUC, and F1-Score and arrive at best model

---

6. A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?

Steps:

1. Data Collection - Collect data containing user behaviour and article related data from news app.

2. Content based recommendation - use TF-IDF and create content based recommendation based on article related data.

3. Collebarative based recommendation - Make a User similarity matrix or Items similarity matrix using Cosine Similarity.

4. Blend both recommenders using a weighted average of similarity scores and recommend articles for users

---

7. You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?

Steps:

1. Data Collecting - Convert all the emails to TF-IDF vectors.

2. Feature Selection - Apply feature selection using chi-square test to select words most correlated with the label (spam or not spam). Optionally use embedded models like Lasso to reduce features.

3. Model Evaluation - Train different classification models using selected features.

4. Best Model Selection - Evaluate using Accuracy, ROC-AUC, and F1-Score and arrive at best model

---

8. An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?

Steps:

1. Data Collection - Collect data containing user ratings, user id and course id

2. User-Item matrix - make a User-Item matrix with ratings as values.

3. Similarity matrix - Make a User similarity matrix or Items similarity matrix using Cosine Similarity.

4. Unique Courses - Identify the courses, based on similar users, unused by the target user and Recommend this.

9.You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?

Steps:

1. Data Preprocessing - Cleaning, handling missing data and scale data.

2. Feature Reduction - Remove highly correlated Features. Then use RFE or SelectKBest and find best Features needed. Alternatively you can also apply PCA to reduce Dimensionality

3. Model Evaluation - Train different models using selected features.

4. Best Model Selection - Evaluate using RMSE for regression model and arrive at best model

---

10. How do you recommend products to new users who haven't interacted with anything yet?

Steps:

1. Use Popularity based Recommendation - based on highly rated or mostly purchased products.

2. Use Content based Recommendation - If age / location / gender / any other details provided by user.